RadarMOT: Radar-Informed 3D Multi-Object Tracking under Adverse Conditions

Bingxue Xu[1] Ajinkya Khoche[2] Patric Jensfelt[1]

[1] KTH Royal Institute of Technology, Sweden [2] Scania/Traton AB, Sweden

***Abstract*—The challenge of 3D multi-object tracking is achieving robustness in real-world applications, for example under adverse conditions and maintaining consistency as distance increases. To overcome these challenges, sensor fusion approaches that combine LiDAR, cameras, and radar have emerged. However, existing multimodal methods usually treat radar as another learned feature inside the network. When the overall model degrades in difficult environments, the robustness advantages that radar could provide are also reduced. In this paper we propose RadarMOT, a radar-informed 3D multi-object tracking framework that explicitly uses radar point clouds as additional observations to refine state estimation and recover objects missed by the detector at long ranges. Evaluations on the MAN-TruckScenes dataset show that RadarMOT consistently improves the Average Multi-Object Tracking Accuracy (AMOTA) by 12.7% at long range and up to 10.3% in adverse weather. The code will be available at https://github.com/bingxue-xu/radarmot**

***Index Terms*—3D multi-object tracking, radar association, autonomous driving, adverse conditions**

I. Introduction

Safe 3D perception requires a system to continuously understand *what* objects are around it and *where they are going* before acting. This is the task of 3D multi-object tracking (3D MOT): maintaining identities and motion trajectories of detected objects across time. Trajectory prediction and motion planning depend directly on the quality of these estimates. 3D MOT operates by detecting objects independently in each frame and then associating them across time to form trajectories. This makes 3D MOT performance fundamentally dependent on two things: (i) *how accurately motion is estimated* (state estimation accuracy), and (ii) *how reliably detections are matched with tracked objects* (data association robustness). LiDAR and cameras are the dominant perception sensors and have driven rapid progress in 3D detection and 3D MOT [1]–[5]. However, both struggle at long range and under adverse conditions. LiDAR point clouds become sparse at long range (see Fig. 1 detection stage) and degrade in fog, rain, and snow. Camera-based detectors show larger 3D depth errors at longer ranges [2] and are sensitive to light [6]. On the other hand, radar is a motion-rich but geometry-poor sensor. Radar directly measures objects' radial velocity via the Doppler effect, providing instantaneous motion without frame-to-frame differencing. It also works reliably in poor weather,

Corresponding author's e-mail: bingxue@kth.se

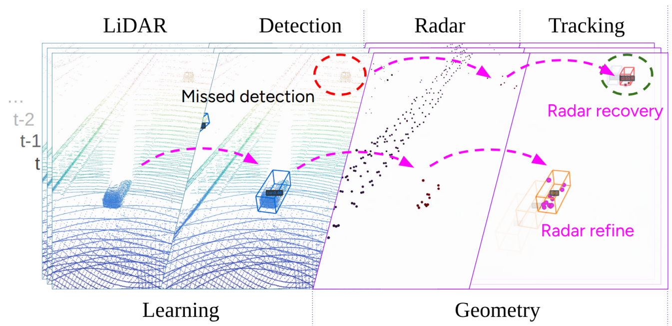

Fig. 1: When state estimation becomes unreliable under adverse conditions, radar radial velocity can help refine object states and recover missed detections, especially at long range.

low light, and at long range [6], making it a good complement to LiDAR and cameras where they are weakest.

Recent radar fusion methods [7]–[10] tend toward data-driven, mostly early-fusion strategies that learn radar features within the detection network. This is mainly due to: 1) detection primarily determining overall tracking performance, and 2) radar point clouds are sparse and noisy. Multimodal fusion methods [8], [11]–[13] explicitly use radar-refined detection velocity in data association to improve tracking performance. Despite achieving strong results on tracking benchmarks, these feature-fusion approaches still depend primarily on geometric cues from LiDAR or semantic cues from images. When these two sensors struggle under adverse weather, the fused representations also become unreliable, and the models undermine radar's inherent advantages.

To reduce the computation burden on the GPU, we revisit both the radar sensing mechanism and Kalman filter-based 3D MOT frameworks [14]–[18]. We first incorporate radar point clouds as an additional measurement in the Kalman filter update (see Fig. 1 radar refine), which consistently boosts Average Multi-Object Tracking Accuracy (AMOTA) and lowers false positives, indicating a more accurate state estimation. Next, we address motion blur during data preprocessing by leveraging radial velocity. Based on the observation that the distance-based association [1] yields more true positives than IoU-based method [15], yet at the cost of more identity switches, we further address the mismatch problem in a two-stage scheme that utilizes bidirectional distances with radar association (see Fig. 1 radar recovery), thereby stabilizing the tracking system.

Our proposed method RadarMOT achieves consistent improvements on the MAN-TruckScenes dataset [19], demonstrating that explicitly leveraging radar physical measurements in the tracking stage can substantially increase robustness when LiDAR and cameras degrade. The contributions of this paper are:

- We propose **RadarMOT**, a radar-informed 3D MOT framework that integrates radar as an explicit observation for tracking without deep learning, demostrates robustness under adverse conditions and at long ranges.
- We introduce a practical **motion compensation** pipeline for multi-sweep radar aggregation that accounts for both ego motion and target motion, compensates temporal offsets using radar Doppler measurements.
- We formulate a **radar-informed Kalman filter update** that uses radar radial velocities to refine the estimated state of tracked objects.
- We propose a **two-stage association** strategy with cross-check association followed by radar association to reduce mismatches and recover objects missed by detector, particularly at long range.

II. RELATED WORK

3D MOT is dominated by the geometry-based Tracking-by-Detection (TBD) paradigm and the learning-based Tracking-by-Attention (TBA) paradigm. Most radar-based multimodal fusion approaches fall within the TBA category.

Tracking by Detection TBD divides the tracking task into four independently modules: detection, motion model, data association, and lifecycle management [17]. AB3DMOT [14] established a strong baseline by coupling a Kalman filter [20] with Intersection over Union (IoU)-based Hungarian matching [21]. The following methods have progressively refined the association strategy, from generalized IoU to normalization [22], lifting it to 3D [17], and combining distance with IoU [15]. Another representative approach, CenterPoint [1], replaces IoU with the Euclidean distance combined with greedy matching, achieving strong performance on nuScenes [23] and Waymo [24]. [16] replaces the Euclidean distance with the Mahalanobis distance. Other methods [18], [25]–[27] enhanced the motion model within the Kalman filter. SimpleTrack [17] proposes a two-stage association. MCTrack [15], consolidates these advances into a unified and general framework that refines the Kalman filter and incorporates both IoU and distance into the data association stage, achieving state-of-the-art performance across the KITTI [28], nuScenes [23], and Waymo [24] datasets. We therefore adopt MCTrack as our baseline.

Tracking by Attention Emerging TBA approaches replace explicit associations with learned queries that carry object identity across frames. Recent 3D MOT methods [11], [29]–[33] encode tracked objects as queries, which attend to features of the current frame to both update object states and detect new objects in a single step, eliminating the need for hand-crafted cost metrics or lifecycle heuristics. Early approaches [29],

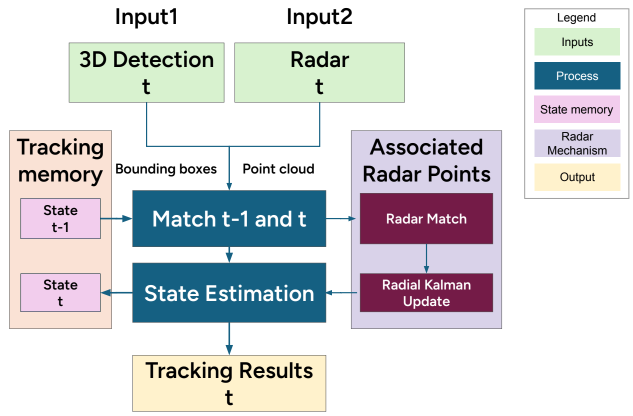

Fig. 2: Overall framework of the proposed RadarMOT, built upon MCTrack [15] and incorporating radar. It first associates tracked objects from both detection and radar in two stages, and then refines the state estimate by a radial velocity informed Kalman filter.

[34] introduced the joint detection-and-tracking framework by propagating object queries over time and preserving identities via attention. Following work [35] extended this idea to longer temporal modeling through track-aware supervision and improved training schemes. Subsequent MOTR [35] series skip the intermediate output detection and introduce end-to-end tracking frameworks. Despite these advances, a common weakness of TBA methods is tight coupling. If the detector or features degrade (bad weather, night, long range, sparse points), tracking can degrade too, and changing sensors usually means retraining.

Radar-based Tracking Radar has recently regained attention due to its robustness to adverse weather, particularly in the scope of camera-radar fusion [36], [37]. A dominant direction incorporates radar at the detection stage before tracking [8]–[10], [38] and improves 3D MOT performance by radar-enhanced detection. References [11], [12], [39] train models using radar velocity as input and employ the predicted bounding box velocity in motion-aware association. DopplerTrack [40] revisits Doppler velocity for moving object tracking, highlighting the potential of radar for motion-centric tracking. RaTrack [39] presents a radar-only method for tracking moving objects that avoids relying on 3D bounding boxes and object classification by using a multi-task network. Similarly, our approach performs motion segmentation, but we extract moving objects via geometric filtering, thus bypassing the need for deep learning.

III. METHOD

We propose RadarMOT, as shown in Fig. 2. It builds upon MCTrack [15] and incorporates radar data. We treat radar data as additional observations alongside 3D detection bounding boxes. The radial velocity is used to refine state estimation in the Kalman filter, and the radar points indicate object presence when objects are temporarily missed by the detector.

This direct design increases the robustness of tracking when detection is weak and compatible with any detector.

A. Motion Compensation

We observed the motion blur problem caused by the ego motion and target motion while aggregating multi-sweep radar data, and addressed this displacement using both ego motion and radar radial velocity.

Ego motion compensation Following the ego-motion compensation [19], we observe that static objects still have residual radial velocities when the vehicle rotate. Therefore we also account for the ego rotation. By doing so, we correctly remove both the translational and rotational ego-motion effects. Let i denote the ith radar point, $\boldsymbol{b}_i$ denotes the line-of-sight unit vector of radar point p_i. After ego motion compensation, the point velocity in the reference coordinate system is written as:

$$\boldsymbol{v}_{\text{comp}}^{(i)} = \boldsymbol{v}_{\text{rel}}^{(i)} + \boldsymbol{v}_{\text{ego}} + \boldsymbol{\omega}_{\text{ego}} \times \boldsymbol{r}^{(i)} \tag{1}$$

where $\boldsymbol{v}_{\text{rel}}^{(i)}$ denotes the radar-reported relative velocity of point i, $\boldsymbol{v}_{\text{ego}}$ is the ego linear velocity, $\boldsymbol{\omega}_{\text{ego}}$ is the ego angular velocity, and $\boldsymbol{r}^{(i)}$ is the vector from the ego rotation center to the radar point.

Radar motion compensation After ego-motion compensation, dynamic points can still appear displaced because radar points are acquired at the sweep timestamp rather than at the keyframe timestamp. Unlike LiDAR methods that infer object motion via scene flow [41], radar Doppler directly measures velocity, capturing non-ego motion without learning. To mitigate the temporal offset, each radar point is translated according to its estimated motion over the time interval between the sweep timestamp t_s and the keyframe timestamp t_k. The resulting motion-compensated point position is given by:

$$\boldsymbol{p}_{\text{comp}}^{(i)} \approx \boldsymbol{p}_{k}^{(i)} + \Delta r^{(i)} \boldsymbol{b}_{i}^{(i)} \tag{2}$$

where $\boldsymbol{p}_{k}^{(i)}$ denotes the point position after transformation to the keyframe in the reference coordinate system, $\boldsymbol{b}_{i}^{(i)}$ is the radar line-of-sight unit vector, and $\Delta r^{(i)} = v_{\text{radial}}^{(i)}(t_k - t_s)$ is the estimated radial displacement during the time offset.

Fig. 3 shows the effect of the proposed motion compensation in the bird's-eye view (BEV). Because radar measures only radial velocity, tangential velocity is unobservable, causing residual distortion, especially for objects crossing the radar's line of sight. This effect is strongest near the ego vehicle, so a nearby region is discarded (15 m ego-centric radius in this implementation).

B. Radar Informed Kalman filter

To update state estimation with radar information, we treat the associated radar radial velocity as an additional observation in a Kalman filter, thereby refining the tracked objects' planar velocity and indirectly stabilizing their positions over time.

In short, for each tracked object we (i) associate radar points that fall inside an inflated track box and have consistent line-of-sight velocity with the box motion (4), (ii) build an observation matrix $\boldsymbol{H}_{r,k}$ whose rows project the track's planar velocity onto each radar point's line-of-sight direction (5)(7), and (iii) update the state with the innovation between measured and predicted radial velocities (9). This uses radar primarily to constrain velocity, which improves tracking robustness under occlusion and reduces velocity drift.

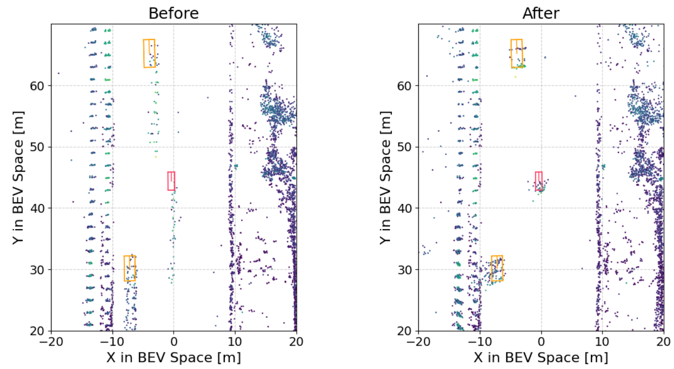

Fig. 3: Comparison of radar point clouds before and after motion compensation. Bounding boxes indicate annotated ground truth.

Motion Model We use a constant-velocity motion model, Let k denote the keyframe index. The state $\mathbf{x}_k = [x_k, y_k, v_{x,k}, v_{y,k}]^\top$ denotes the object position and planar velocity in the global coordinate system. F denotes the state transition matrix, $\boldsymbol{Q}_k$ denotes the process noise. The predicted state is then given as:

$$\boldsymbol{x}_{k|k-1} = \boldsymbol{F}\boldsymbol{x}_{k-1|k-1} + \boldsymbol{w}_k \qquad \boldsymbol{w}_k \sim \mathcal{N}(\mathbf{0}, \boldsymbol{Q}_k) \tag{3}$$

Radar Observation Model To handle noisy radar points, only radar points likely to be reflecting from the same object are used as observations. Let p_i denote the i-th radar point, and let $\boldsymbol{p}_s^{(i)} \in \mathbb{R}^3$ denote its position, with measured radial velocity $v_{\text{radar}}^{(i)} \in \mathbb{R}$, and let $v_{\text{box}}^{(i)} \in \mathbb{R}$ denote the bounding box velocity projected onto the line-of-sight of point i. The associated radar point set is defined by two filters:

$$\mathcal{R}_{\text{assoc}} = \left\{ p_i \in \mathcal{R}_k \;\middle|\; \boldsymbol{p}_g^{(i)} \in \alpha\mathcal{B}_k^t \;\wedge\; |v_{\text{radar}}^{(i)} - v_{\text{box}}^{(i)}| \le \delta_v \right\} \tag{4}$$

where

- $\alpha\mathcal{B}_k$ restricts radar points to the track bounding box in XY plane, α denotes the inflation rate.
- δ_v removes outliers based on the radial velocity difference between each radar point $v_{\text{radar}}^{(i)}$ and the bounding box velocity $v_{\text{box}}^{(i)}$ projected onto the radar point's line-of-sight direction.

To construct the radar observation matrix $\boldsymbol{H}_{r,k}$, where r means observation from radar at the k-th frame, the line-of-sight unit vector $\boldsymbol{b}_i$ of each associated radar point i is defined in the radar sensor coordinate system S_i as:

$$\boldsymbol{b}_i = \frac{\boldsymbol{p}_s^{(i)}}{\left\|\boldsymbol{p}_s^{(i)}\right\|} \in \mathbb{R}^3 \tag{5}$$

To project the global-frame track velocity into the radar sensor frame S_i, we apply the rotation ${}^{S_i}\boldsymbol{R}_{\text{global}}$ and retain

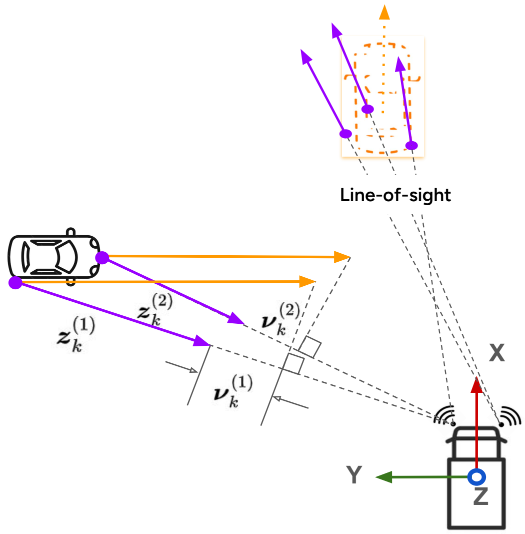

Fig. 4: Radar observation and radial velocity-informed Kalman update. Purple points and arrows represent radar points and motion-compensated velocity. Orange arrows represent the predicted velocity of the tracked object.

only the planar (XY) rows. Since radar measures only radial velocity, the observation function for p_i projects the track velocity onto the line-of-sight direction, position entries of $\boldsymbol{h}_i^\top$ are zero. The observation function h_i for p_i is then:

$$h_i(\boldsymbol{x}_k) = \boldsymbol{h}_i^\top \boldsymbol{x}_k, \qquad \boldsymbol{h}_i^\top = \begin{bmatrix} 0 & 0 & \boldsymbol{b}_i^{\top\, S_i} \boldsymbol{R}_{\text{global},xy} \end{bmatrix} \in \mathbb{R}^{1\times 4} \tag{6}$$

Stacking M such rows gives the joint observation matrix:

$$\boldsymbol{H}_{r,k} = \begin{bmatrix} \boldsymbol{h}_1^\top \\ \vdots \\ \boldsymbol{h}_M^\top \end{bmatrix} \in \mathbb{R}^{M\times 4} \tag{7}$$

Let $\boldsymbol{z}_k^r = [z_1, z_2, \ldots, z_M]^\top \in \mathbb{R}^M$ denote the vector of radial velocities measured by the M associated radar points. Following the standard linear observation model [42], the joint radar observation model is as:

$$\boldsymbol{z}_k^r = \boldsymbol{H}_{r,k}\boldsymbol{x}_k + \boldsymbol{n}_k, \qquad \boldsymbol{n}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_{r,k}) \tag{8}$$

where $\boldsymbol{x}_k$ is the true object state, and $\boldsymbol{R}_{r,k} \in \mathbb{R}^{M\times M}$ denotes the measurement noise covariance.

Radar-Informed Kalman Filter Update Given the radar observation, the tracking state is then updated using a Kalman filter, as illustrated in Fig. 4, the radar-informed state is updated by:

$$\begin{aligned} \boldsymbol{\nu}_k &= \boldsymbol{z}_k^r - \boldsymbol{H}_{r,k}\boldsymbol{x}_{k|k-1} \\ \boldsymbol{x}_{k|k} &= \boldsymbol{x}_{k|k-1} + \boldsymbol{K}_k\boldsymbol{\nu}_k \end{aligned} \tag{9}$$

where

- $\boldsymbol{\nu}_k$ denote the innovation derived from the difference between the observed radial velocity and the motion model predicted radial velocity.
- $\boldsymbol{z}_k^r$ denotes the observed radial velocity.
- $\boldsymbol{H}_{r,k}\boldsymbol{x}_{k|k-1}$ denotes the track state estimate predicted radial velocity.
- $\boldsymbol{x}_{k|k-1}$ and $\boldsymbol{x}_{k|k}$ denote the predicted and updated track states.
- $\boldsymbol{K}_k$ is the Kalman gain.

C. Two-Stage Association

To reduce mismatches, which are the primary cause of identity switches in tracked objects, we propose a two-stage association that first performs a bidirectional cross-check association and then applies a radar association.

Cross-Check Association Instead of associating at a single timestamp (e.g., either comparing the forward prediction with the detection at time t, or comparing the track with the backward predicted detection at time $t-1$), we use both forward and backward predictions for mutual cross-checking and combine a speed similarity to further minimize the influence of object orientation. The cost function is defined as:

$$cost = \frac{\alpha\|\boldsymbol{p}_t^{trk} - \boldsymbol{p}_t^{det}\|}{v_t^{trk}\Delta t} + \frac{\beta\|\boldsymbol{p}_{t-1}^{trk} - \boldsymbol{p}_{t-1}^{det}\|}{v_t^{det}\Delta t} + \frac{\delta|v^{trk} - v^{det}|}{\max(v^{trk}, v^{det})} \tag{10}$$

where

- $\boldsymbol{p}_t^{trk}$ and $\boldsymbol{p}_t^{det}$ are the track and detection centers at time t.
- $\boldsymbol{p}_{t-1}^{trk}$ and $\boldsymbol{p}_{t-1}^{det}$ are the track center and backward-predicted detection center at time $t-1$.
- v^{det} and v^{trk} are the detection and track speeds, reducing the influence of orientation error.
- α, β, and δ are weighting factors. Since these terms provide complementary cues for robust association, all weights are uniformly set to 0.5 to balance their contributions.

Radar Association To recover the objects missed by the detector, unmatched tracked objects are associated with radar measurements in a second stage.

We first remove the points that are close to detections to avoid duplicate associations and remove stationary clutter, then select candidate radar points for unmatched tracked objects by position and velocity similarity.

$$\mathcal{R}_{\text{assoc}}^t = \left\{ p_i \in \mathcal{R}_k \;\middle|\; p_i \in \beta\hat{\mathcal{B}}_k^t \;\wedge\; |v_{\text{radar}}^{(i)} - v_{\text{pred}}^{(i)}| \le \delta_v \right\} \tag{11}$$

where $\beta\hat{\mathcal{B}}_k^t$ restricts radar points to the track *predicted* bounding box, β denotes the inflation rate. δ_v removes outliers based on the radial velocity difference between each radar point $v_{\text{radar}}^{(i)}$ and the predicted track velocity $v_{\text{pred}}^{(i)}$ projected onto the radar point's line-of-sight direction.

If more than $N_{\min} = 3$ points are collected, the track is considered radar-seen for the current frame, and the radar-informed Kalman filter update (9) is applied to refine the state estimation.

IV. EXPERIMENTS SETUP

Datasets Experiments are conducted on the largest public radar dataset MAN-TruckScenes dataset (TruckScenes) [19]

TABLE I: Performance comparison on MAN-TruckScenes validation set with the same detector. (Bic., Motor, Ped., Tra., Tru.) denote (Bicycle, Motorcycle, Pedestrian, Trailer, Truck). Bold indicates best performance per column.

Method	**Detector**	**AMOTA % ↑**								**TP ↑**	**FP ↓**	**FN ↓**	**IDS ↓**
		Overall	**Bic.**	**Bus**	**Car**	**Motor**	**Ped.**	**Tra.**	**Tru.**				
CenterPoint [1]	CenterPoint [1]	22.8	9.3	0.0	51.1	23.5	37.7	0.9	37.2	37149	**8467**	26912	7278
MCTrack [15]	CenterPoint [1]	26.6	11.9	0.0	40.0	**59.4**	36.4	0.7	37.6	33636	13026	32378	5325
RadarMOT (ours)	CenterPoint [1]	**33.3**	**17.0**	0.0	**53.6**	56.2	**46.5**	**13.0**	**46.8**	**42717**	12257	**24906**	**3716**

for autonomous driving, which includes 747 scenes of 20 seconds each, with keyframes at 2 Hz. The sensor suite includes 4 cameras, 6 LiDAR units, and 6 4D radar sensors with full 360-degree coverage. It covers diverse weather conditions (rain, fog, snow), lighting situations (darkness, glare, twilight), and driving areas (highway, urban, rural). This dataset also reflects the dynamic, long-range perception and occlusion challenges: 40% of annotated objects move faster than 20 m/s, 50% of all annotations are beyond 75 m, and 39% of the objects have a visibility of less than 41% [19]. Thus, evaluation is conducted using this dataset to investigate robustness.

Evaluation Metrics We follow the official nuScenes tracking challenge and report AMOTA. For all metrics, true positives are obtained by applying a 2 m threshold to the 2D center distance on the ground plane. AMOTA and MOTAR are defined as:

$$\text{AMOTA} = \frac{1}{n-1} \sum_{r \in \left\{\frac{1}{n-1}, \frac{2}{n-1}, \ldots, 1\right\}} \text{MOTAR} \tag{12}$$

$$\text{MOTAR} = \max\left(0,\ 1 - \frac{\text{IDS}_r + \text{FP}_r + \text{FN}_r - (1-r)\cdot P}{r \cdot P}\right) \tag{13}$$

where n denotes the number of recall sampling points, and r is the recall threshold. P refers to the number of ground-truth positives for the current class. (IDS,FP,FN) are the number of (identity switches, false positives, false negatives).

As this is the first 3D MOT work on TruckScenes, the classes follow nuScenes tracking challenges [23] and ranges are adopted from TruckScenes detection benchmark [19], i.e., bigger vehicles (car, truck, bus, trailer) are evaluated up to 150 m and smaller objects (pedestrian, motorcycle, bicycle) up to 75 m.

Detection Input We use the same 3D detection results obtained by CenterPoint [1] on the TruckScenes validation dataset. The detection model is reproduced following the training recipe in TruckScenes [19]. It's worth mentioning that CenterPoint [1] achieves 58% mean Average Precision (mAP) on nuScenes but only 33% mAP on TruckScenes validation set, indicating that the environmental conditions in TruckScenes are more challenging for 3D perception.

Baseline We use MCTrack [15] as baseline, as described in Section II. We report the MCTrack tracking results on the TruckScenes validation set in the second row of Table I. Additionally, we report the CenterPoint tracker in the first row of Table I.

TABLE II: Comparison across ranges on TruckScenes validation set. All methods use the same CenterPoint [1] detector. Bold indicates best value per column.

Method	**Overall**	**I: 0–50 m**	**II: 50–100 m**	**III: 100–150 m**
	AMOTA % / IDS	**AMOTA % / IDS**	**AMOTA % / IDS**	**AMOTA % / IDS**
CenterPoint [1]	22.8 / 7278	28.1 / 2766	21.5 / 3107	21.2 / 1518
MCTrack [15]	26.6 / 5325	30.8 / 1804	23.8 / 1704	20.1 / **836**
RadarMOT (ours)	**33.3 / 3716**	**36.0 / 1321**	**32.2 / 1451**	**32.8** / 895

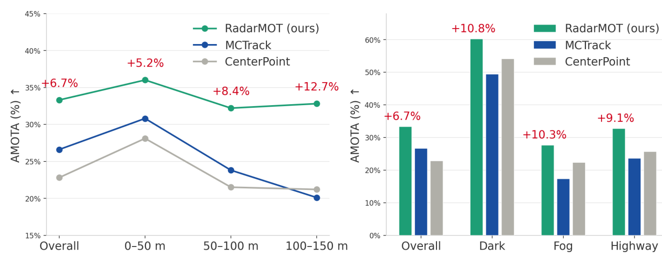

Fig. 5: Comparison across range and adverse conditions on TruckScenes validation set.

V. Results

We report our results on the TruckScenes validation set in Table I. Our proposed method RadarMOT achieves 33.3% in terms of overall AMOTA, an absolute improvement of 6.7% over the MCTrack baseline, and a relative 30% reduction in Identity Switches (IDS). We found that the distance-based association method is preferable in highly dynamic scenarios. We observed that CenterPoint achieves more true positives and fewer false negatives than MCTrack, but at the cost of a substantially increased number of IDS. Thanks to the bidirectional distance association and radar-refined state estimation, RadarMOT combines the strengths of both methods without introducing additional false positives. To the best of our knowledge, this is the first 3D MOT investigation on the TruckScenes, offering a conservative baseline for the community.

Long Range To reveal how RadarMOT behaves as range increases, we evaluate the performance across three range bins up to 150 m, as shown in Table II and Fig. 5. The growing margin reflects radar's increasing contribution as LiDAR point density drops with range. Our method achieves an absolute +12.7% AMOTA improvement at the third bin (100–150 m) compared to baseline MCTrack.

Adverse Conditions We also evaluate the performance under adverse conditions in different weather, lighting, and driving

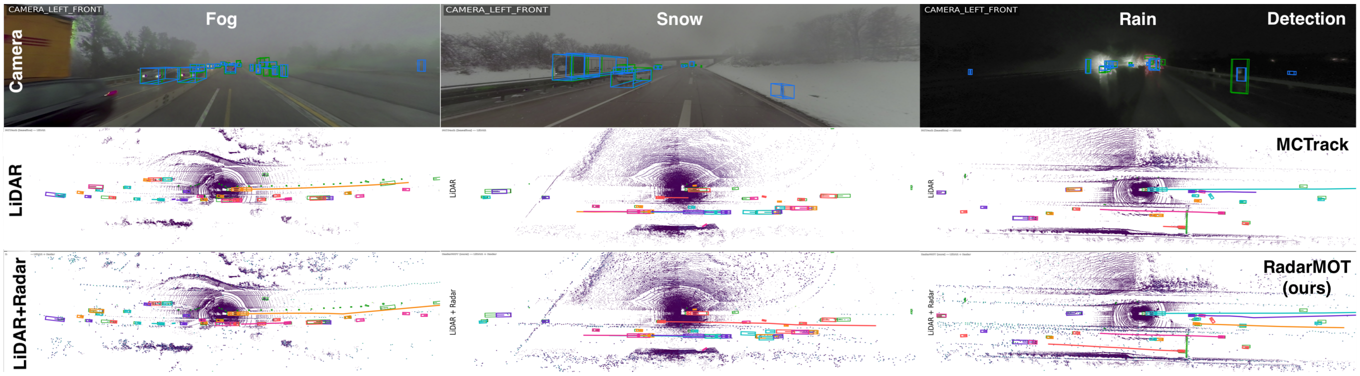

Fig. 6: Qualitative comparison on the TruckScenes dataset in adverse weather conditions at distances up to 150 m. Images are shown for context. The second and third rows present the tracking results of the baseline MCTrack [15] and RadarMOT (ours), respectively. Green bounding boxes denote the ground truth, and the colored bounding boxes and trajectory lines correspond to tracked objects.

TABLE III: Comparison under adverse weather, lighting conditions and driving areas on TruckScenes validation set. Bold indicates the best value per row per metric.

Group	Condition	CenterPoint [1]		MCTrack [15]		RadarMOT (ours)	
		AMOTA%	IDS	AMOTA%	IDS	AMOTA%	IDS
Weather	Clear	36.0	3040	41.4	2454	**46.0**	**1552**
	Overcast	24.2	2753	29.0	1775	**33.6**	**1307**
	Rain	20.9	1458	18.3	839	**26.1**	**771**
	Fog	22.3	173	17.3	109	**27.6**	**76**
	Snow	16.2	38	**20.0**	34	17.8	**25**
Lighting	Dark	54.1	1388	49.4	674	**60.2**	728
	Illuminated	22.7	4627	33.0	2952	**38.8**	**2317**
	Twilight	25.8	1095	26.8	906	**31.0**	**508**
	Glare	16.3	494	13.2	361	**16.8**	**219**
Area	Highway	25.7	5833	23.6	3975	**32.7**	**2702**
	City	21.5	887	24.4	687	**27.4**	**577**
	Rural	16.4	463	19.1	377	**23.3**	**303**
	Residential	25.3	53	**29.6**	83	28.6	**54**
	Terminal	51.9	33	56.5	**9**	**56.6**	13

areas, see Table III and Fig. 5. RadarMOT achieves consistent absolute AMOTA improvements over the baseline MCTrack: +10.3% in fog, +10.8% at night, and +9.1% on highways. Contrary to our initial hypothesis, RadarMOT performs worse than the baseline in snow.

Qualitative Rerults Fig. 6 illustrates the robustness of our approach compared to the baseline MCTrack [15] under foggy, snowy, and rainy night conditions. A higher number and longer durations of trajectories (colored lines) indicate that RadarMOT is able to track more objects and maintain their tracks more consistently.

Ablation Study To isolate the contribution of each component, we add them incrementally on the baseline MCTrack [15], as shown in Table IV. Radar Kalman filter refinement achieves the lowest false positives and reduces IDS. Adding radar association captures more true positives but also increases false positives. This trade-off is stabilized by the cross-check association, which boosts additional true positives, highlighting that robust association is key before refinement.

TABLE IV: Ablation study on TruckScenes validation set. Each row adds one component on top of the MCTrack [15] baseline. Bold indicates best results.

Component			AMOTA%↑	TP↑	FP↓	FN↓	IDS↓
Radar Kalman filter	Radar Association	Cross-Check Association					
			26.6	33636	13026	32378	5325
✓			29.2	33948	12126	32234	5157
✓	✓		30.7	38446	12861	26515	6378
✓	✓	✓	**33.3**	**42717**	**12257**	**24906**	**3716**

VI. Conclusion

We present an explicit radar fusion method for 3D MOT which does not depend on deep learning. It integrates radar data as additional observations and performs robustly under adverse conditions and at long ranges. As the first 3D MOT study conducted on the TruckScenes dataset, we release our code to offer a solid baseline for the community. Thanks to the raw data fusion strategy, RadarMOT is practically suitable for robotics and autonomous driving applications.

References

[1] T. Yin, X. Zhou, and P. Krahenbuhl, "Center-based 3d object detection and tracking," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 11 784–11 793.

[2] X. Jiang, S. Li, Y. Liu, S. Wang, F. Jia, T. Wang, L. Han, and X. Zhang, "Far3d: Expanding the horizon for surround-view 3d object detection," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 38, no. 3, 2024, pp. 2561–2569.

[3] Y. Liu, T. Wang, X. Zhang, and J. Sun, "Petr: Position embedding transformation for multi-view 3d object detection," in *European conference on computer vision*. Springer, 2022, pp. 531–548.

[4] S. Wang, Y. Liu, T. Wang, Y. Li, and X. Zhang, "Exploring object-centric temporal modeling for efficient multi-view 3d object detection," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 3621–3631.

[5] Y. Wang, V. C. Guizilini, T. Zhang, Y. Wang, H. Zhao, and J. Solomon, "Detr3d: 3d object detection from multi-view images via 3d-to-2d queries," in *Conference on robot learning*. PMLR, 2022, pp. 180–191.

[6] X. Huang, Z. Xu, H. Wu, J. Wang, Q. Xia, Y. Xia, J. Li, K. Gao, C. Wen, and C. Wang, "L4dr: Lidar-4dradar fusion for weather-robust 3d object detection," 2025. [Online]. Available: https://arxiv.org/abs/2408.03677

[7] R. Nabati and H. Qi, "Centerfusion: Center-based radar and camera fusion for 3d object detection," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2021, pp. 1527–1536.

[8] P. Wolters, J. Gilg, T. Teepe, F. Herzog, A. Laouichi, M. Hofmann, and G. Rigoll, "Unleashing hydra: Hybrid fusion, depth consistency and radar for unified 3d perception," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 7467–7474.

[9] K. Lei, Z. Chen, S. Jia, and X. Zhang, "Hvdetfusion: A simple and robust camera-radar fusion framework," *arXiv preprint arXiv:2307.11323*, 2023.

[10] C. Zhang, H. Wang, L. Chen, Y. Li, and Y. Cai, "Mixedfusion: An efficient multimodal data fusion framework for 3-d object detection and tracking," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 1, pp. 1842–1856, 2023.

[11] N. Baumann, M. Baumgartner, E. Ghignone, J. Kühne, T. Fischer, Y.-H. Yang, M. Pollefeys, and M. Magno, "Cr3dt: Camera-radar fusion for 3d detection and tracking," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 4926–4933.

[12] Y. Li, J. Deng, Y. Xiao, Y. Wang, X. Chu, J. Ji, and Y. Zhang, "Farfusion v2: A geometry-based radar-camera fusion method on the ground for roadside far-range 3d object detection," in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 8421–8430.

[13] P. Wolters, J. Gilg, T. Teepe, F. Herzog, F. Fent, and G. Rigoll, "Sparc: Sparse radar-camera fusion for 3d object detection," *arXiv preprint arXiv:2411.19860*, 2024.

[14] X. Weng, J. Wang, D. Held, and K. Kitani, "Ab3dmot: A baseline for 3d multi-object tracking and new evaluation metrics," *arXiv preprint arXiv:2008.08063*, 2020.

[15] X. Wang, S. Qi, J. Zhao, H. Zhou, S. Zhang, G. Wang, K. Tu, S. Guo, J. Zhao, J. Li *et al.*, "Mctrack: A unified 3d multi-object tracking framework for autonomous driving," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 4551–4558.

[16] H.-k. Chiu, J. Li, R. Ambruş, and J. Bohg, "Probabilistic 3d multi-modal, multi-object tracking for autonomous driving," in *2021 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2021, pp. 14 227–14 233.

[17] Z. Pang, Z. Li, and N. Wang, "Simpletrack: Understanding and rethinking 3d multi-object tracking," in *European conference on computer vision*. Springer, 2022, pp. 680–696.

[18] P. Zhang, X. Li, X. Lin, and L. He, "Easy-poly: A easy polyhedral framework for 3d multi-object tracking," *arXiv preprint arXiv:2502.17822*, 2025.

[19] F. Fent, F. Kuttenreich, F. Ruch, F. Rizwin, S. Juergens, L. Lechermann, C. Nissler, A. Perl, U. Voll, M. Yan *et al.*, "Man truckscenes: A multimodal dataset for autonomous trucking in diverse conditions," *Advances in Neural Information Processing Systems*, vol. 37, pp. 62 062–62 082, 2024.

[20] K. Ramachandra, *Kalman filtering techniques for radar tracking*. CRC Press, 2018.

[21] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

[22] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-iou loss: Faster and better learning for bounding box regression," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, no. 07, 2020, pp. 12 993–13 000.

[23] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuscenes: A multimodal dataset for autonomous driving," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 11 621–11 631.

[24] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine *et al.*, "Scalability in perception for autonomous driving: Waymo open dataset," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 2446–2454.

[25] X. Li, T. Xie, D. Liu, J. Gao, K. Dai, Z. Jiang, L. Zhao, and K. Wang, "Poly-mot: A polyhedral framework for 3d multi-object tracking. in 2023 ieee," in *RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 9391–9398.

[26] X. Li, D. Liu, Y. Wu, X. Wu, L. Zhao, and J. Gao, "Fast-poly: A fast polyhedral framework for 3d multi-object tracking," *arXiv preprint arXiv:2403.13443*, 2024.

[27] X. Liu, X. Zhao, G. Liu, Z. Wu, T. Wang, L. Meng, and Y. Wang, "Imm-mot: A novel 3d multi-object tracking framework with interacting multiple model filter," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 20 540–20 547.

[28] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? the kitti vision benchmark suite," in *2012 IEEE conference on computer vision and pattern recognition*. IEEE, 2012, pp. 3354–3361.

[29] T. Meinhardt, A. Kirillov, L. Leal-Taixe, and C. Feichtenhofer, "Trackformer: Multi-object tracking with transformers," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 8844–8854.

[30] T. Zhang, X. Chen, Y. Wang, Y. Wang, and H. Zhao, "Mutr3d: A multi-camera tracking framework via 3d-to-2d queries," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 4537–4546.

[31] Z. Pang, J. Li, P. Tokmakov, D. Chen, S. Zagoruyko, and Y.-X. Wang, "Standing between past and future: Spatio-temporal modeling for multi-camera 3d multi-object tracking," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 17 928–17 938.

[32] M. Y. Lee, C. D. W. Lee, J. Li, and M. H. Ang, "Dino-mot: 3d multi-object tracking with visual foundation model for pedestrian re-identification using visual memory mechanism," *IEEE Robotics and Automation Letters*, vol. 10, no. 2, pp. 1202–1208, 2024.

[33] C. Zhang, C. Zhang, Y. Guo, L. Chen, and M. Happold, "Motiontrack: End-to-end transformer-based multi-object tracking with lidar-camera fusion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, June 2023, pp. 151–160.

[34] P. Sun, J. Cao, Y. Jiang, R. Zhang, E. Xie, Z. Yuan, C. Wang, and P. Luo, "Transtrack: Multiple object tracking with transformer," *arXiv preprint arXiv:2012.15460*, 2020.

[35] F. Zeng, B. Dong, Y. Zhang, T. Wang, X. Zhang, and Y. Wei, "Motr: End-to-end multiple-object tracking with transformer," in *European Conference on Computer Vision (ECCV)*, 2022.

[36] S. Yao, R. Guan, Z. Peng, C. Xu, Y. Shi, W. Ding, E. Gee Lim, Y. Yue, H. Seo, K. Lok Man, J. Ma, X. Zhu, and Y. Yue, "Exploring radar data representations in autonomous driving: A comprehensive review," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 6, pp. 7401–7425, 2025.

[37] K. Shi, S. He, Z. Shi, A. Chen, Z. Xiong, J. Chen, and J. Luo, "Radar and camera fusion for object detection and tracking: A comprehensive survey," *IEEE Communications Surveys Tutorials*, vol. 28, pp. 3478–3520, 2026.

[38] R. Nabati, L. Harris, and H. Qi, "Cftrack: Center-based radar and camera fusion for 3d multi-object tracking," in *2021 IEEE Intelligent Vehicles Symposium Workshops (IV Workshops)*, 2021, pp. 243–248.

[39] Z. Pan, F. Ding, H. Zhong, and C. X. Lu, "Ratrack: Moving object detection and tracking with 4d radar point cloud," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 4480–4487.

[40] Y. Zeng, T. Wu, S. Qi, J. Li, X. Duan, and Y. Yu, "Simpler is better: Revisiting doppler velocity for enhanced moving object tracking with fmcw lidar," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 7842–7849.

[41] Q. Zhang, A. Khoche, Y. Yang, L. Ling, S. S. Mansouri, O. Andersson, and P. Jensfelt, "Himo: High-speed objects motion compensation in point clouds," *IEEE Transactions on Robotics*, vol. 41, pp. 5896–5911, 2025.

[42] S. Thrun, W. Burgard, and D. Fox, *Probabilistic robotics*. Cambridge, Mass.: MIT Press, 2005.